

EDUCATION

- 2026.03–Present** **Ph.D. Student in Data Science**
Graduate School of Data Science, Kyungpook National University, Korea
Advisor: Prof. Suhyeon Kim
- 2024.03–2026.02** **M.S. in Data Science**
Graduate School of Data Science, Kyungpook National University, Korea
Dissertation: *A Newtonian-informed Transformer AutoEncoder for Vessel Navigation Risk Index Estimation*
Advisor: Prof. Suhyeon Kim
- 2017.03–2024.02** **B.S. in Mathematics**
Pusan National University, Korea
- 2022.03–2024.02** **Interdisciplinary Major in Industrial Mathematics & Software**
Pusan National University, Korea
- 2023.03–2023.12** **Undergraduate Researcher**
Industrial Mathematics Center, Pusan National University, Korea

RESEARCH INTERESTS

- AI for mobility, industrial systems, and medical data
- Federated learning, privacy-preserving AI, and cybersecurity
- Risk estimation, anomaly detection, and decision-support modeling
- Graph machine learning and spatiotemporal representation learning

PUBLICATIONS

Notation. [†] equal contribution; * corresponding author.

SCI(E) / SSCI Journal Papers

- [1] Eun-ji Lee[†], **Byeongkeun Kwon**[†], Suhyeon Kim*, Sung Hyun Noh*, “Predictive Modelling of the Need for Tracheostomy after Traumatic Cervical Spinal Cord Injury using Machine Learning,” *Yonsei Medical Journal*, 2026. Co-first author; JCR Q1.
- [2] **Byeongkeun Kwon**[†], Dain Lee[†], Hyunjun Ko, Hanbin Lee, Hyeonjun Hwang*, Suhyeon Kim*, “A Graph Neural Network Approach for Aquatic Biodiversity Prediction Leveraging Water System Interconnections,” *Ecological Informatics*, May 2026. Co-first author; JCR Q1, 4.1%.

Scopus Papers

- [1] **Byeongkeun Kwon**, Suhyeon Kim*, “Driver Behavior Anomaly Detection based on Federated Learning Considering Data Distribution Imbalance,” *The Journal of Korean Institute of Communications and Information Sciences*, 50(3), 395–405, March 2025. First author; Best Paper of JKICS.

Working Papers

- [1] **Byeongkeun Kwon**, Dain Lee, Suhyeon Kim*, “A Newtonian-informed Transformer Autoencoder for Fishing Vessel Navigation Risk Index Estimation.” Under review at *Ocean Engineering*.
- [2] **Byeongkeun Kwon**[†], Dasom Seong[†], Jiyun Park, Hyeonjun Hwang*, Suhyeon Kim*, “Water Quality Risk Index Estimation using Graph Neural Networks for Spatiotemporal Watershed Systems.” Submitted to *Water Research*.

- [3] **Byeongkeun Kwon**[†], Dasom Seong[†], Sung Hyun Noh^{*}, Suhyeon Kim^{*}, “SSIU-Net: Spine Screw Inverse U-Net for Anatomical Representation-guided Pedicle Screw Trajectory Planning.” Manuscript in preparation. Target journal: *Medical Image Analysis*.
- [4] **Byeongkeun Kwon**[†], Sesun You[†], Young Seop Son^{*}, Suhyeon Kim^{*}, “CAN-based Heavy-Duty Truck Weight Estimation: A Dual-phase Analysis of AI and Vehicle Dynamics.” Manuscript in preparation. Target journal: *IEEE Transactions on Intelligent Vehicles*.

CONFERENCE PRESENTATIONS

International Conferences

1. **Byeongkeun Kwon**, Dasom Seong, Suhyeon Kim, Sung Hyun Noh, “3D Generative Models for Vertebral Segmentation and Pedicle Screw Path Prediction,” *MICCAI 2025 CLINICCAI*, Daejeon, Korea, September 23–27, 2025.
2. **Byeongkeun Kwon**, Eun-ji Lee, Suhyeon Kim, Sung Hyun Noh, “Development of Machine Learning Model Predicting the Need for Tracheostomy after Cervical Spinal Cord Injury,” *ASIA SPINE & NCS 2025*, Busan, Korea, September 4–6, 2025.
3. Hyosung Kim, Taejun Kwon, **Byeongkeun Kwon**, Jieun Jang, Jihun Jang, “DRIVE: Drowsiness Recognition and Intelligent Vibration Evocation,” *IEEE World Haptics Conference 2025*, Suwon, Korea, July 8–11, 2025.
4. **Byeongkeun Kwon**, Suhyeon Kim, “Classical Mechanics Meet Deep Learning: Ship Hazardous Navigation Detection System,” *IISE Annual Conference & Expo 2025*, Atlanta, USA, May 31–June 3, 2025. *Presenter; Session Chair*.
5. **Byeongkeun Kwon**, Suhyeon Kim, “Baseball Match Prediction based on Graph Neural Network,” *Deep Learn 2024: 11th International School on Deep Learning*, Porto, Portugal, July 15–19, 2024.

Domestic Conferences

1. **Byeongkeun Kwon**, Dasom Seong, Suhyeon Kim, “Graph Neural Network-based Water Quality Risk Index for Spatio-Temporal Watershed Systems,” *Proceedings of KIIE 2026 Joint Conference*, Gyeongju, Korea, June 4–6, 2026.
2. **Byeongkeun Kwon**, Seseun You, Dain Lee, Dasom Seong, Suhyeon Kim, Hyung Jong Cho, Hyuntae Cho, Young Seop Son, Suhyeon Kim, “Lightweight MLP-Mixer for Real-time Weight-in-Motion with Embedded Deployment,” *Proceedings of KSAE 2026 Spring Conference*, Jeju, Korea, May 20–23, 2026.
3. **Byeongkeun Kwon**, Dasom Seong, Suhyeon Kim, Sung Hyun Noh, “3D Generative Models for Vertebral Segmentation and Pedicle Screw Path Prediction,” *The 44th Annual Spring Meeting of the Korean Neurosurgical Society*, Jeju, Korea, April 16–18, 2026.
4. **Byeongkeun Kwon**, Dain Lee, Dasom Seong, Suhyeon Kim, “Development of a Generative Adversarial Networks-based Fuzzing Test Technique for Vehicle Communication Security,” *Proceedings of KSAE 2025 Fall Conference*, Busan, Korea, November 12–15, 2025.
5. **Byeongkeun Kwon**, Suhyeon Kim, “A Newtonian-informed Transformer AutoEncoder for Vessel Navigation Risk Estimation,” *Proceedings of KIIE 2025 Fall Conference*, Daejeon, Korea, November 6–7, 2025.
6. **Byeongkeun Kwon**, Suhyeon Kim, “Mixer-integrated Graph Neural Networks for Accurate and Efficient Prediction of Crystalline Solid Properties,” *Proceedings of KIIE 2025 Joint Conference*, Jeju, Korea, June 18–21, 2025. *Poster*.
7. **Byeongkeun Kwon**, Suhyeon Kim, “Privacy-preserving Federated Learning for Abnormal Driving Behavior Detection,” *Proceedings of KICS Fall Conference 2024*, Gyeongju, Korea, November 20–22, 2024.
8. Eun-ji Lee, **Byeongkeun Kwon**, Suhyeon Kim, Sung Hyun Noh, “Development of Machine Learning Model Predicting the Need for Tracheostomy after Cervical Spinal Cord Injury,” *NCS 2024*, Seoul, Korea, September 5–7, 2024.
9. **Byeongkeun Kwon**, Suhyeon Kim, “Baseball Match Prediction based on Graph Neural Network,” *Proceedings of Korean Artificial Intelligence Association 2024 Conference*, Busan, Korea, August 16–17, 2024. *Poster*.

AWARDS AND HONORS

Grants

2026 Spring **KNU Ph.D. Fellow**, Kyungpook National University. Award amount: KRW 5,000,000 (approximately USD 3.3K).

Awards

2025 **Minister of Science and ICT Award**, 2025 K-Data Science Conference & Hackathon, Hackathon Track.

2025 **Minister of Science and ICT Award**, 2025 ICT Challenge.

2025 **Minister of Environment Award**, 2025 Environmental Data Utilization & Analysis Competition, Analysis Track.

2024 **President Award of NIA**, 2024 K-Data Science Conference & Hackathon, Hackathon Track.

2024 **Chairperson Award of Korea National Park Service**, 2024 Environmental Data Utilization & Analysis Competition, Analysis Track.

2024 **Finalist Award**, 2024 Weather Big Data Contest, Fog Track.

2024 **Finalist Award**, 2024 Weather Big Data Contest, Power Track.

2023 **Finalist Award**, 2023 Weather Big Data Contest.

2023 **Encouragement Award**, 2023 Yeongnam Mathematical Society, Poster Presentation.

2023 **Excellence Award & Encouragement Award**, 2023 Mathematics May Competition, Departmental Contest.

PATENTS

1. Aquatic biodiversity prediction system and its method, 2025.

TEACHING ASSISTANTSHIPS

2026 Spring **Teaching Assistant**
Data Science Deep Learning (DASC0735), Graduate School of Data Science, Kyungpook National University

2026 Spring **Teaching Assistant**
Graph Data Analysis (DASC0767), Graduate School of Data Science, Kyungpook National University

ACADEMIC SERVICE

2025 Session Chair, *IISE Annual Conference & Expo 2025*, Atlanta, USA.

sectionResearch Experience and Projects

Ongoing Projects

2026.03–2031.02 **Development of Graph Deep Learning Core Technologies for Hypertrustworthy Dynamic Structural AI Countering Cybersecurity Threats**
National Research Foundation of Korea, Individual Basic Research / Excellent Young Researcher Type B
Researcher

2026.04–2028.03	Advancement of an Adaptive AI-Based Industrial Control Protocol Security Testing Automation Platform and Expansion into Industrial Domains Industry-Academia-Research Collabo R&D Program, Commercialization R&D Program	Researcher
2025.03–2027.08	BK21 FOUR Project: GLOW-AI Innovative Talent Development Education and Research Group BK21 FOUR	Researcher
2023.06–2026.05	Sequential Data Analysis using Multi-view Graph Deep Representation Learning and Its Industrial Applications National Research Foundation of Korea, Creative Challenge Research Support Program	Researcher
2024.07–Present	Data-driven Intelligent Mobility Research Center IITP, ICT Research Center / ITRC Program	Researcher
2023.09–Present	Regional Demand Oriented Data Science Human Resource Development for Regional Industry Innovation Data Science Convergence Human Resource Development Program	Researcher

Completed Projects

2025.05–2025.12	Generative AI Platform for Automated Analysis of Industrial Control Protocol Specifications and Automated Security Vulnerability Test-case Generation Industry-Academia-Research Collabo R&D Program, Preliminary Research Program	Researcher
2024.08–	GAN-based Automated Test-case Generation and Security Vulnerability Detection for Vehicle Communication Protocols and Infotainment File Formats Korean Ministry of SMEs and Startups, Startup Growth Technology Development Program	Researcher

REFERENCES

Prof. Dr. Suhyeon Kim | Kyungpook National University
 Assistant Professor
 Graduate School of Data Science
 Bldg. 4th Joint Lecture Hall, 101-5, 80, Daehak-ro, Buk-gu, Daegu 41566
 Republic of Korea
 Email: suhyeonkim@knu.ac.kr
 Webpage: <https://sites.google.com/view/suhyeon-kim/home>